

A GUIDE TO PRODUCT QUALITY STANDARDS GLASSWARE



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ABBREVIATIONS

<i>APLAC</i>	Asia Pacific Laboratory Accreditation Co-operation
<i>ANSI</i>	American National Standards Institute
<i>AS</i>	Australian Standards
<i>ASTM</i>	American Society for Testing and Materials
<i>BIS</i>	Bureau of Indian Standards
<i>CE</i>	Conformité Européenne
<i>CoC</i>	Certificate of Conformity
<i>CPSC</i>	Consumer Product Safety Commission
<i>CPSIA</i>	Consumer Product Safety Improvement Act
<i>CSPA</i>	Consumer Product Safety Act
<i>EC</i>	Council Regulation (Regulation of the European Parliament and Council)
<i>EFSA</i>	European Food Safety Authority
<i>EN</i>	European Standards
<i>EU</i>	European Union
<i>FCM</i>	Food Contact Materials
<i>FDA</i>	Food and Drug Administration
<i>FHSA</i>	Federal Hazardous Substances Act
<i>FSSAI</i>	Food Safety and Standards Authority of India
<i>GCC</i>	Gulf Co-operation Council
<i>GSO</i>	GCC Standardization Organization
<i>IS</i>	Indian Standard
<i>ISI</i>	Indian Standards Institution
<i>ILAC</i>	International Laboratory Accreditation Co-operation
<i>ISO</i>	International Organization for Standardization
<i>MLA</i>	Multilateral Recognition Arrangement
<i>MRA</i>	Mutual Recognition Arrangement
<i>MSME</i>	Micro, Small & Medium Enterprises
<i>NABCB</i>	National Accreditation Board for Certification Bodies
<i>NABL</i>	National Accreditation Board for Testing and Calibration Laboratories
<i>RoHS</i>	Restriction of Hazardous Substances Directive
<i>SASO</i>	Saudi Standards, Metrology and Quality Organization
<i>QCI</i>	Quality Council of India
<i>QCO</i>	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2,40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and

Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

Glassware industry has been identified as one of the key industries with glass products being an important item in the array of goods targeted at exports from Uttar Pradesh.

As Uttar Pradesh has abundant reserves of Silica sand, it is one of the leading states in glass production in India. Firozabad is the largest producer having over 400 small units involved in manufacturing of bangles, glassware and glass sculptures. Firozabad has earned the name 'Suhag Nagri' for its high-quality bangles, which were its most famous product. With the government incentivizing the industry, many units have been set up which manufacture variety of glassware and related products using both traditional as well as industrial techniques.

GLASS BANGLES, GLASS BOTTLES, KITCHEN WARE AND DECORATIVE GLASS ARTICLES (FIROZABAD)

1. INTRODUCTION

Firozabad has been producing glass bangles for more than 200 years now and is the biggest manufacturer of glass bangles in the world. Artisans engaged in the manufacturing of glassware use a special traditional technique called 'Mouth-Blowing' for making various products such as lanterns, Christmas trees, kitchenware and various decorative items. Approximately 20,000 artisans make a variety of glassware in the district.



Glassware comprises of any decorative article made of glass, often designed for everyday use. From very early times, glass has been used for various kinds of vessels, and in all countries where the industry has been developed, glass has been produced in a great variety of forms and kinds of decoration, much of it of great beauty.

Bangle-making is a household business in the district with traditional technique being passed on through

generations. It is also called 'Suhag Nagari' because of the bangles.

For the purpose of this compendium, following products have been considered:

- Bangles
- Kitchenware
- Bottles
- Decorative Items

The glassware of Firozabad is renowned the world over and have been given a GI tag in 2014, which further enhances their presence in the global markets as a unique product.

2. RAW MATERIALS

Glass is made from natural and abundant raw materials. At high temperature, glass is structurally similar to liquids, however, at ambient temperature it behaves like solids. The key ingredients for glass making are:

- Sand
- Soda ash
- Limestone
- Clarifying agents
- Colours/dyes
- Glistening glass

3. MANUFACTURING PROCESS

3.1. Glass Bangles

Bangles manufactured here are the ones with a joint and each bangle is joined manually. Different kind of tools are involved in making glass

bangles as most of them are custom made by the artisans. The manufacturing process of bangles is as under:

- a. Bangles are made out of soda glass melted in a tank furnace with temperatures around 1300 °C to 1400 °C. A tank furnace is essentially a large box of refractory material holding a quantity of glass.
- b. The artisan uses a long iron pole to scoop out a glob of molten glass from the tank furnace and then swiftly transfers it to another artisan who shapes the glob into a conical shape by rotating the rod and with the help of a customized tool which looks like a trowel.
- c. Further, it is transferred to another furnace where the artisan works in synchronization with the machine and draws a thin filament of glass from the melt and places it steadily on the rotating rod which is constantly rotated on a motor.
- d. The molten glass loops around the rod taking the shape of a bangle. The thickness is controlled by exerting the requisite amount of pull on the softened glass and these rotating rods come in different diameters depending on size of the bangle which is required to be made.

Before the molten glob exhausts, next glob is attached to the previous one to keep continuity.

- e. Another artisan who is on the other end of the furnace uses a metallic ruler/ pointed tool which prevents bangles from sticking to each other on the spindle and also makes sure that only one layer of glass is rolled on. When the spindle gets filled up, it is removed and the glass bangle is taken out, which looks like a long glass spiral.
- f. These spirals are then cut with the help of a diamond cutter, which separates each bangle from another. The separated bangles which are open ended and not complete are then taken to join them over a candle or a kerosene lamp which is called 'Judai' which literally means joining.
- g. The bangles are melt slightly to join the ends, a process usually performed by women artisans/workers. Bangles are further decorated with zari, silver/golden powder in different intricate patterns and designs on the surface.
- h. The bangles are then finally baked in a furnace called 'pakai bhatti' which smoothens the sharp edges and makes the bangles bright and attractive.

- i. Bangles are intertwined, strung together and packed in cardboard boxes to be transported, exported or sold in the market.

3.2. Glass Bottles and Kitchen Ware

Glass is chemically inert and more impermeable than plastic, making it withstand better the processes like pasteurization while containing more corrosive contents. Glass bottles are especially crucial in the food and beverage packaging, since they do not affect the flavour of its content and prevent loss of carbonation better than the plastic counterparts. Glass bottles have multiple uses including holding beer and wine, glass milk bottles, baby bottles, soda bottles, and water bottles. Outside of beverages, glass bottles & containers are used to hold food ingredients, sauces and jams. Additionally, glass is used to package products like medication, cosmetics, and essential oils.

The first stage of the glass bottle production process begins with the hot end processes, which typically employ high amounts of heat to produce and shape a glass container.

- a. A furnace is first used to mould molten glass, which is fed to the furnace as glass feedstock. Soda-lime glass stock accounts for the majority (around 90 percent) of the types of glass

products, and is typically largely comprised of silica, with about 10 percent each of calcium oxide and lime. Small amounts of aluminium oxide, ferric oxide, barium oxide, sulphur trioxide, and magnesia also account for about 5 percent of soda-lime glass. Before melting, cullet (recycled glass) is added to the stock, accounting for anywhere between 15 and 50 percent of the final glass composition.

- b. Additional elements to colour the glass are also added before the glass is melted. Amber glass bottles get their colour from nickel, sulphur, and carbon, although natural impurities in sand can also cause colouration. Other colours can be achieved through adding various materials.

Once the stock is fed into the furnace, temperatures inside can be as high as 1200 °C. The molten glass is then fed through a refiner, which allows air bubbles to escape the liquid. Next, one of the following forming methods is applied: press-and-blow or blow-and-blow.

- **Press-and-Blow Moulding Glass:** Press-and-blow formation takes place in an Individual Section (IS) machine and is the more commonly used method in

glass-container production. IS machines have between five and twenty sections, all identical, which can each carry out the glass-container forming process simultaneously and completely. The result is that five to twenty containers can be produced with one machine at the same time.

When the molten glass reaches between 1050 °C and 1200 °C it is said to be in its plastic stage, and it is during this phase that press-and-blow formation begins. A shearing blade is used to cut and shape the glass into a

cylindrical shape, called a gob. The cut gob falls and using gravitational force, rolls through the appropriate passage to reach the mould. A metal plunger presses the gob into the blank mould, where it assumes the mould's shape and is then termed a "parison". Next, the parison is moved into a final mould, where it is blown into the mould to assume its final dimensions. This process is typically used for wide-mouthed glass containers, but can also be used to manufacture thin-necked bottles.

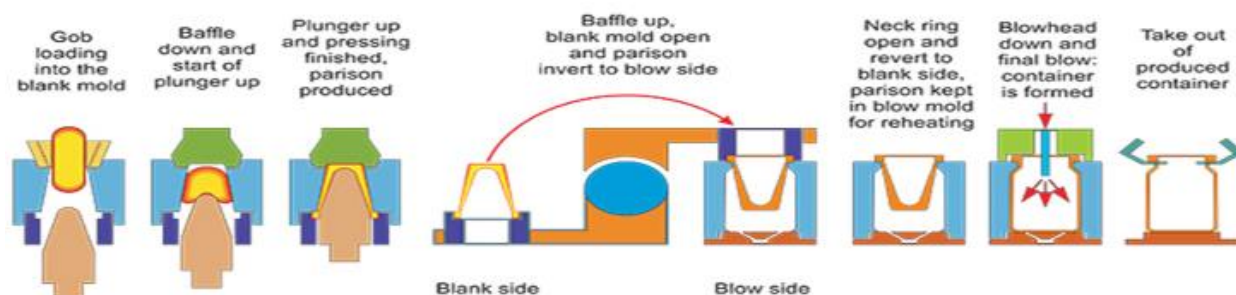


Figure 1: Press and blow glass moulding process

- **Blow-and-Blow:** Like press-and-blow formation, blow-and-blow takes place in an IS machine, where a gob is released during the plastic stage and moved along to the moulds. However, in blow-and-blow formation, the gob is forced into the blank mould using compressed air to push

the gob into place. The gob, now a parison, is flipped into a corresponding final mould where it is blown again, to form the interior side of the glass container. Glass bottles of varying neck thickness can be made using blow-and-blow formation.

After formation, bottles often undergo internal treatment, a process which makes the inside of the bottle more chemically-resistant, an important factor, if the bottles are intended to hold alcohol or other degrading substances. Internal

treatment can take place during formation or directly after and typically involves treating the bottles with a gas mixture of fluorocarbon. Glass containers can also be treated externally to strengthen the surface or reduce surface friction.

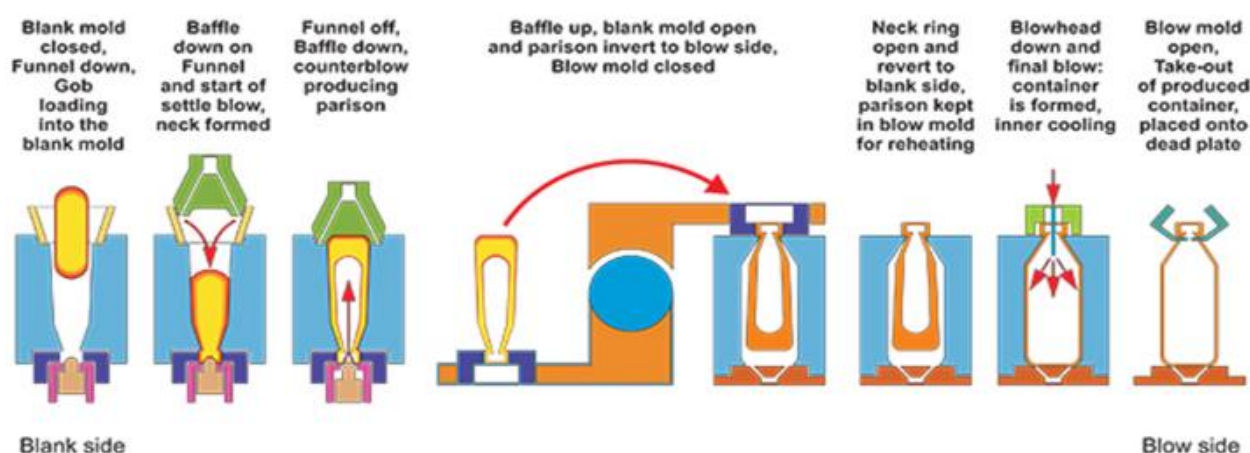


Figure 2: Blow and blow glass moulding process

- c. **Annealing Glass Bottles:** Once formation is complete, some bottles may suffer from stress as a result of unequal cooling rates. An annealing oven, also known as a lehr, can be used to reheat and cool glass containers to rectify stress and make the bottle stronger.
- d. **Cold End Processes:** At this stage in glass production, the bottles or glass containers are inspected and packaged. Inspection is often done by a combination of automated and mechanical inspection, to ensure integrity of the final product.

Common faults include checks (cracks in the glass) and stones (pieces of the furnace that melt off and are subsequently worked into the final container), which are important to identify because they can compromise the component. Packaging methods vary from factory to factory depending on the specific type of bottle and the size of the production run.

Amber glass bottles are especially popular for medication, beer, and other contents that are sensitive to light. The amber colour of the bottle filters out UV and blue light, allowing it to

prevent the breakdown of these products, called photo oxidation. Without this filtering of higher-energy light waves, beer may break down over time and ruin its flavour, while essential oils may shift in scent.

3.3. Decorative Glass Articles

- a. **Glass blowing:** A combination of sand and soda lime, along with colouring agents is used to produce the raw material which is melted together to form molten glass. This glass is “gathered” from the furnace onto a blowpipe – glassblowers use air blown into the pipe combined with movements of the glass, and special tools, to quickly shape the molten glass before it cools. Once properly shaped, the art glass sculpture is placed in an annealing oven to slowly cool down the glass until it is stable. If it cools too quickly, the glass art may break or crack. Sometimes cracks in existing glass sculptures can be the result of improper annealing time or incompatibility between the colours within the glass sculptures.
- b. **Hot Sculpting:** This technique is used when a solid metal rod gathers the molten glass from the furnace and it is shaped with the use of special tools. While the

process is similar to blown glass, no actual blowing takes place in the sculpture. The hot sculpting process tends to make much larger solid pieces of glass art.

- c. **Cold Working:** This method of glass work involves working with glass while it's in its cold or annealed state. The artist uses a variety of techniques, including sandblasting, cutting, grinding, polishing and engraving to create a unique piece of art. Cold working also includes the application of special glues to adhere glass pieces together. These techniques are especially important for glass art repair and restoration

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check for air bubbles or micro cracks, if any, on the surface
- Check for chipped edges at the mouth of the bottles
- Check for sharp edges

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers, e.g., REACH Regulations of European Union (EU), CCC mark of China, etc. Regulations, domestic and international (some key

regions/countries), are provided in the tables below. However, other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)

- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	India	a. Food Safety and Standards (Licensing and Registration of Food Businesses) Regulations, 2011.	a. Food Safety and Standards Authority India (FSSAI)	a. Governs use of food grade utensils apart from other food safety standards b. All equipment and utensils which come in contact with food.

5.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	European Union	a. The Materials and Articles in Contact with Food (England) Regulations, 2012 b. Food safety – Heavy metals in Ceramic, Glass and Enamelled table and Kitchenware c. Regulation (EC) 2023/2006 d. Regulation (EC) 1935/2004	a. European commission	a. Specifies requirements with respect to chemicals and heavy metals b. Specifies good manufacturing practices for materials coming in contact with food c. GMP for food contact materials (FCM). d. Certification that the food contact material (FCM) shall not release constituents into food.
2	United States of America	a. Consumer Product Safety Improvement Act (CPSIA), Act of 2008 b. Consumer Product Safety Act (CPSA) Act of 1972	a. California State Consumer products safety commission b. Food and Drug Administration (FDA)	a. Specifies requirements with respect to chemicals and heavy metals b. Specifies requirement for suitable product labelling c. Specifies banned substances d. Any manufacturing unit having more than ten

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
		c. Federal Hazardous Substances Act (FHSA), Act of 1960. d. California Proposition 65, Act of 1986. e. Food and Drug Administration Regulations		employees are required to label products if they contain carcinogenic products
3	Australia	a. Environmental & Chemical Regulations	a. Department of Agriculture, Water and the Environment	a. Lead in stained glass b. Lead canes and solders used in stained glass and lead lighting can be a health hazard if lead dust is swallowed or inhaled. c. Even occasional craftspeople are at risk.
4	Middle East	a. GSO Technical Regulations - GSO 2231: 2012 - GSO 839/1997	a. SABER b. Saudi Standards, Metrology and Quality Organization (SASO) c. Ministerial committee for food safety of the GCC countries	a. Specifies technical regulations for conformity as per SASO regulations for food contact materials.

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/Country	Standard	Aspects Covered
General (Refer Table 1)	IS 1382: 1981 (Reaffirmed Year: 2018)	Glossary of terms relating to glass and glassware	Australia	AS/NZS 4668: 2000	Glossary of terms used in the glass and glazing industry
	IS 6654: 1992 (Reaffirmed Year: 2018)	Glossary of terms relating to glass container	USA	ASTM C162 – 05 (2015)	Standard terminology of glass and glass products
	-	-	European Union	BS ISO 7348: 1992	Glass containers. Manufacture: Vocabulary
Raw Materials (Refer Table 2a. & 2.b)	IS 488: 1980 (Reaffirmed Year: 2018)	Specification for glass making sands	USA	ASTM C146 - 21	Standard Test Methods for Chemical Analysis of Glass Sand
	IS 997: 1973 (Reaffirmed Year: 2018)	Specification for limestone and dolomite for glass industry	USA	ASTM C 429 -21	Standard Test Method for Sieve Analysis of Raw Materials for Glass Manufacture

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 1106: 1986 (Reaffirmed Year: 2017)	Specification for distilled water glass bottles	USA	ISO 6486: 2019	Ceramic ware, glass ceramic ware and glass dinnerware in contact with food-
	IS 1107: 1986 (Reaffirmed Year: 2017)	Specification for aerated water glass bottles crown finish type	USA	ISO 7086: 2019	Glass hollowware in contact with food
	IS 1392: 1999 (Reaffirmed Year: 2018)	Glass milk bottles - Specification	USA	ISO 24117: 2020	Tableware, giftware, jewellery and luminaries, made of glass — Glass clarity — Classification and test method
	IS 1961: 1968 (Reaffirmed Year: 2016)	Specification for glass tableware	European Union	BS 6748: 1986	Specification for limits of metal release from ceramic ware, glassware, glass ceramic ware and vitreous enamel ware
	-	-	European Union	BS 6748: 1986	Specification for limits of metal release from ceramic ware, glassware, glass ceramic ware and vitreous enamel ware
Finishing (Refer Table 4.a & 4.b)	IS 2303: Part1: Section 1: 2021	Method of grading glass for alkalinity	USA	ISO 12818: 2013	Glass packaging — Standard tolerances for flaconnage
	IS 9806: 2001 (Reaffirmed Year: 2017)	Methods of test for and permissible limits of toxic materials released from ceramic ware, vitreous enamelware, glassware and glass- ceramic	USA	ASTM C148 - 17	Standard test methods for polariscopic examination of glass containers
	IS 11930: 2018	Methods of thermal shock tests for glass containers	European Union	BS EN ISO 7458:2004	Glass containers. Internal pressure resistance. Test methods

6.1. Table 1: Domestic & International General Standards

Stage	Code	Scope
General	IS 1382: 1981 (Reaffirmed Year: 2018) - Glossary of terms relating to glass and glassware	- This standard covers the terms and definition relating to glass, glassware and related products - does not cover terms related to glass containers and their defects
	IS 6654: 1992 (Reaffirmed Year: 2018) - Glossary of terms relating to glass container	Defines terms relating to glass containers and their manufacture. Terms relating to closures are specifically excluded

Stage	Code	Scope
	ASTM C162 - 05(2015) - Standard terminology of glass and glass products	Glossary of terms
	BS ISO 7348: 1992 - Glass containers - Manufacture: Vocabulary	Glossary of terms
	AS/NZS 4668 – 2000 - Glossary of terms used in the glass and glazing industry	Glossary of terms

6.2. Table 2.a: Domestic Standards for Raw Materials

Stage	Code	Scope
Raw Material	IS 488: 1980 (Reaffirmed Year: 2018) - Specification for glass making sands	- Prescribes requirements and methods of sampling and test for silica sand suitable for the manufacture of some of the important types of glassware - This standard does not cover silica sand for optical and other special glasses
	IS 997: 1973 (Reaffirmed Year: 2018) - Specification for limestone and dolomite for glass industry	- This standard prescribes requirements and methods of sampling and test for limestone and dolomite for glass industry for making colourless glass - Inferior grades of the material containing, higher percentage of iron and used for making certain types of coloured glasses are not covered in this standard

6.3. Table 2.b: International Standards for Raw Materials

Stage	Code	Scope
Raw Material	ASTM C146 – 21 - Standard Test Methods for Chemical Analysis of Glass Sand	- Prescribes requirements and methods of sampling and test to ensure that the chemical composition of the glass sand meets the compositional specification required for this raw material. - Exclusions: This standard does not cover silica sand for optical and other special glasses
	ASTM C 429 -21 - Standard Test Method for Sieve Analysis of Raw Materials for Glass Manufacture	- Sieve analysis of common raw materials for glass manufacture, such as sand, soda-ash, limestone, alkali-alumina silicates, and other granular materials used in glass batch - Exclusions

6.4. Table 3.a: Domestic Standards for Processing

Stage	Code	Scope
Processing	IS 1106: 1986 (Reaffirmed Year: 2017) - Specification for distilled water glass bottles	Lays down the requirements for distilled water glass bottles
	IS 1107: 1986 (Reaffirmed Year: 2017) - Specification for aerated water glass bottles crown finish type	Prescribes the requirements for aerated water glass bottles of crown finish type
	IS 1392: 1999 (Reaffirmed Year: 2018) - Glass milk bottles specification	Prescribes requirements and methods of sampling and test for cylindrical glass bottles for liquid milk other than sterilized milk
	IS 1961: 1968 (Reaffirmed Year: 2016) - Specification for glass tableware	This standard prescribes the requirements and the methods of sampling and test for glass tableware

6.5. Table 3.b: International Standards for Processing

Stage	Code	Scope
Processing	ISO 6486: 2019 - Ceramic ware, glass ceramic ware and glass dinnerware in contact with food	- Part 1 - Specifies a test method for the release of lead and cadmium from ceramic ware, glass ceramic ware and glass dinnerware intended to be used in contact with food, but excluding vitreous and porcelain enamel articles - Part 2 - specifies permissible limits for the release of lead and cadmium from ceramic ware, glass ceramic
	ISO 7086: 2019 - Glass hollowware in contact with food	- Part 1- Specifies a test method for the release of lead and cadmium from glass hollowware that is intended to be used in contact with food - Part 2- specifies permissible limits for the release of lead and cadmium from glass hollowware that is intended to be used in contact with food
	ISO 24117: 2020 - Tableware, giftware, jewellery and luminaries, made of glass Classification and test method	- This document establishes requirements for the use of the glass designations "clear glass" and "ultra-clear glass" for non-coloured glass items according to their clarity and iron content. - It specifies a procedure for measuring the clarity of glass items by means of a spectrophotometer. - This document is applicable to - Mineral glasses - Glass items where a part is not covered by coating or decoration, and is therefore available for sampling.
	BS 6748: 1986 - Specification for limits of metal release from ceramic ware, glassware, glass ceramic ware and vitreous enamel ware	Specification for limits of metal release from ceramic ware, glassware, glass ceramic ware and vitreous enamel ware
	BS EN ISO 9000-1: 2005 - Glass containers — Vacuum lug finishes — Part 1: General	Types of vacuum lug finishes for glass containers for EN ISO 9100-2 to EN ISO 9100-14

6.6. Table 4.a: Domestic Standards for Finishing

Stage	Code	Scope
Finishing	IS 2303: Part1: Section 1: 2021 - Method of grading glass for alkalinity	PART 1 Method of test and classification: Section 1- Hydrolytic Resistance of Glass Grains at 98°C, Section 2: Hydrolytic Resistance of Glass Grains at 121o C
	IS 11930: 2018 - Methods of thermal shock tests for glass containers	Specify test methods for thermal shock resistance and thermal shock endurance in glass containers
	IS 9806: 2001 (Reaffirmed Year: 2017) - Methods of Test for and permissible limits of toxic materials released from ceramic ware, vitreous enamelware, glassware and glass-ceramic	Prescribes the methods of test for and permissible limits of toxic materials released from ceramic ware, vitreous enamelware, glassware and glass-ceramic ware used in preparation, storage, cooking or serving of food and beverages

6.7. Table 4.b: International Standards for Finishing

Stage	Code	Scope
Finishing	ISO 12818: 2013 - Glass packaging Standard tolerances for flaconnage	Specifies the tolerances for the bottles intended to be used for pharmaceutical products, cosmetic and perfumery products and chemical products. The tolerances concerned are brim-full capacity, height, diameter and width, and verticality.
	ASTM C148 – 17 - Standard test methods for polariscopic examination of glass containers	- Determination of relative optical retardation associated with the state of anneal of glass containers - Test methodology - Exclusions
	BS EN ISO 7458:2004 - Glass containers, internal pressure resistance: Test methods	Test methods for internal pressure resistance.

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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 - d. Gulf Countries - <https://sabersaudi.co/services/saso-product-coc/saso-technical-regulation/>
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9. International Standards -
 - a. USA - <https://www.iso.org/ics/81.040.30/x/>
 - b. Europe - <https://shop.bsigroup.com/ProductDetail?pid=000000000030240642>
 - c. Australia: <https://www.standards.org.au/search?q=thread+textile&mode=allwords&sort=relevance>
 - d. Middle East: <https://www.gso.org.sa/store/>
10. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials,

moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network, which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products,

services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/calibration results between countries, which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such

products, various regulations and the standards making bodies Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For examples, BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL):** <https://nabl-india.org/>
- **Bureau of Indian Standards (BIS):** <https://bis.gov.in/>

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying

repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative

consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and would not cause any harm.
- Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, e.g. USFDA

prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the relevant regulatory authorities, as per the international standards.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's

criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral/Mutual Recognition Arrangements (MLA/MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB):
<http://nabcb.qci.org.in/>

APPENDIX II: IMPORTANT INSTITUTIONS

GLASS				
S. No.	Institute	Areas of Expertise	Postal Address	Contact
1	Centre for the Development of Glass Industry, Industrial Area, Jalesar Rd, Firozabad, Uttar Pradesh 283203	- Product and Technology Development Unit - Consultancy - Testing laboratories	Centre for the Development of Glass Industry, Industrial Area, Jalesar Rd, Firozabad, Uttar Pradesh 283203	Email: cdgizbd@gmail.com Phone: 9897021838 Website: http://www.cdgindia.net/
2	Indian Institute of Technology – Banaras Hindu University (IIT - BHU), Department of Ceramic Engineering	Glass & Glass Ceramics	IIT- BHU, Varanasi, Uttar Pradesh 221005	Email: head.cer@iitbhu.ac.in Phone: 9936182124 Website: www.iitbhu.ac.in/dept/cer
3	CSIR-Central Glass & Ceramic Research Institute, 196, Raja S.C. Mullick Road, Kolkata – 700 032 West Bengal, India	- Non-oxide Ceramics - Consultancy - Refractory and Traditional Ceramics - Specialty Glass	CSIR-Central Glass & Ceramic Research Institute, 196, Raja S.C. Mullick Road, Kolkata – 700 032 West Bengal, India	Email: dir_office@cgcri.res.in Phone: +91 33 2473 0957 Website: https://www.cgcri.res.in/
4	CSIR-Central Glass & Ceramic Research Institute, CSIR-CGCRl Khurja Centre G T Road, Khurja 203131 (Uttar Pradesh)	- Consultancy - Training & Demonstration - Testing Laboratories - Trouble shooting	CSIR-Central Glass & Ceramic Research Institute, CSIR-CGCRl Khurja Centre G T Road, Khurja 203131 (Uttar Pradesh)	Email: hstripathi@cgcri.res.in Phone: +91 5738245433 Website: https://www.cgcri.res.in/campus/khurja

